

COMPANY PROFILE

PIE SQUARE
TECHNOLOGIES PVT LTD

Kusunti-13

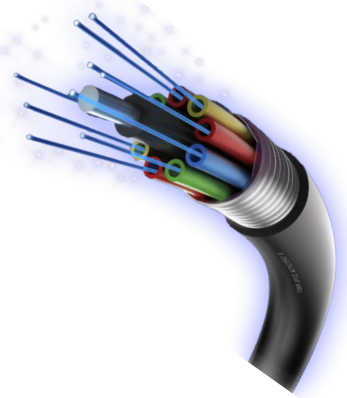
Lalitpur, Nepal

info@piesquaretechnologies.com



About us

Pie Square Technologies Pvt. Ltd. is a pioneering force in Nepal's burgeoning telecommunications sector, established having itself as a company with focus on innovation and excellence, having Registration No. 2071051/075/076 and VAT No. 606587170, in January 2019. Based in Kusunti, Lalitpur, the company operates as a fully privately owned entity, committed to enhancing the nation's technological infrastructure.



Pie Square Technologies addresses the evolving demands of Nepal's digital industry, characterized by developing technological infrastructure. The company plays a crucial role in supporting telecom giants by expanding network coverage through specialized services such as site survey, tower construction and installation, RF drive testing, similarly, assisting Internet provider companies to expand and maintain their optical fiber network. Piesquare not only enhances its technological portfolio but also plays a vital role in fostering a greener future projects like deployment of solar panels, wind turbines, alternative energy systems that utilizes natural resources sustainably. By harnessing local talent and resources, Pie Square Technologies delivers high-quality, technology-driven solutions that bolster connectivity and drive progress across Nepal, reflecting a steadfast dedication to customer satisfaction and national development.



Our Services

Telecom and Fiber Services

We provide a comprehensive array of telecom services, which encompass tower erection, BTS installation, commissioning and dismantle, RF drive test & optimization, as well as optical fiber installation and maintenance.

Solar services

We specialize in the installation of solar power systems to provide clean, renewable energy solutions for our clients.

IT Services

Our IT services encompass delivering different IT related solutions for organizations alongside offering consulting support to assist clients in reaching their technology objectives.

Telecom services

1) Tower construction and Erection

we specialize in the construction and erection of telecom towers, ensuring seamless network expansion for telecom operators across Nepal. Our expertise covers the end-to-end process of building robust, high-stability tower infrastructure that supports reliable mobile, internet, and broadcasting services.

Our Services Include:

- ✓ **Site Survey & Feasibility Study** – Conducting detailed geological, structural, and environmental assessments to determine optimal tower locations.
- ✓ **Foundation & Civil Work** – Constructing strong and durable foundations using industry-standard materials and engineering best practices.
- ✓ **Tower Erection** – Installing various types of telecom towers, including self-supporting, guyed mast, monopole, and rooftop towers.
- ✓ **Antenna & Equipment Installation** – Mounting antennas, microwave dishes, and other telecom equipment with precision for optimal network performance.
- ✓ **Power & Grounding Systems** – Setting up power backup solutions (solar, battery, and generator systems) and proper grounding for safety and efficiency.
- ✓ **Structural Strengthening & Maintenance** – Conducting tower reinforcement, inspections, and periodic maintenance to ensure long-term durability.



Telecom services

2) Tower Installation, Commissioning, Dismantle and Replacement

At Pie Square Technologies Pvt. Ltd., we provide comprehensive installation and commissioning services for telecom equipment, ensuring seamless network performance and expansion. Our expertise covers the deployment of antennas, microwave links, power plants, batteries, rectifiers, and related infrastructure, enabling telecom operators to maintain high-performance and reliable connectivity across their networks.

Our Services Include:

✓ **Site Survey & upgradability Study** – Conducting detailed assessments of existing sites for possible upgrades and replacements.

✓ **Antenna & Microwave Link Installation** – Precise mounting and alignment of 2G, 3G, 4G, and 5G antennas for optimal network coverage.

✓ **Microwave Link Deployment** – Installing and configuring point-to-point and point-to-multipoint microwave links to ensure high-speed data transmission.

✓ **Power Plant Installation & Dismantling** – Setting up power distribution units, backup power systems, and hybrid power solutions for telecom sites, Safely removing and replacing old power plants, ensuring minimal downtime.

✓ **Battery Installation & Dismantling** – Installing backup battery banks, Proper removal and eco-friendly disposal of old telecom batteries following environmental guidelines.

✓ **Rectifier Installation & Dismantling** – Deploying high-efficiency rectifiers to optimize power conversion and reduce energy loss. Safely dismantling outdated rectifier units and replacing them with modern energy-efficient systems.

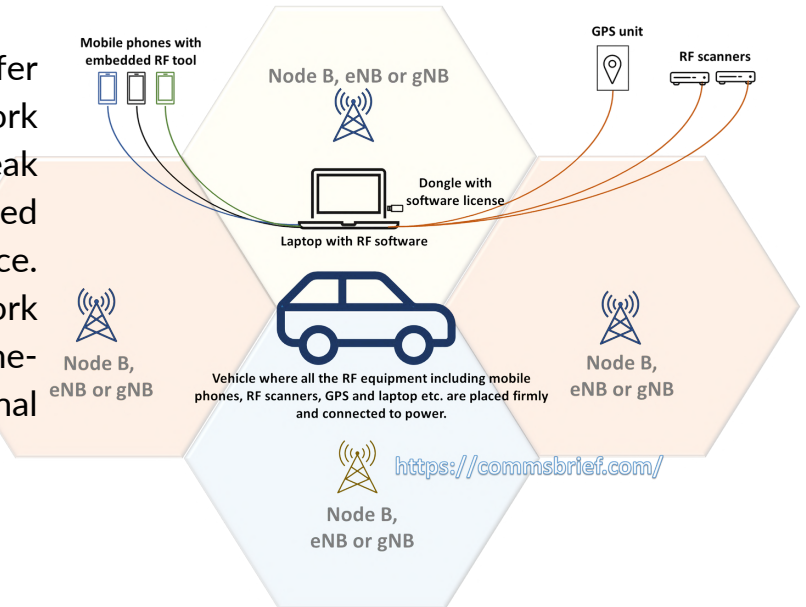
✓ **Structural Reinforcement & Safety Measures** – Ensuring secure installation of telecom antennas and microwave equipment, following all industry standards.



Telecom services

3) RF Drive Test and Optimization

At Pie Square Technologies Pvt. Ltd., we offer extensive RF Drive Testing and Network Optimization services aimed at ensuring peak network performance, uninterrupted connectivity, and an enhanced user experience. Our skilled engineers perform real-time network evaluations, pinpoint coverage gaps, and fine-tune network parameters to boost signal strength, quality, and capacity.



Our Services Include:

RF Drive Test & Analysis

- ✓ **Single Site Verification (SSV)** – Conducting detailed drive tests for newly installed or upgraded cell sites, ensuring proper coverage, signal quality, and handover performance.
- ✓ **Cluster Drive Test** – Performing drive tests for a group of sites, analyzing overall network performance, and identifying weak coverage areas.
- ✓ **Dedicated Walk Testing** – Conducting indoor and outdoor walk tests in key locations such as malls, airports, and commercial buildings for indoor coverage assessment.
- ✓ **Benchmarking Drive Test** – Comparing network performance against competitors to evaluate signal quality, throughput, and coverage strength.



Physical Optimization

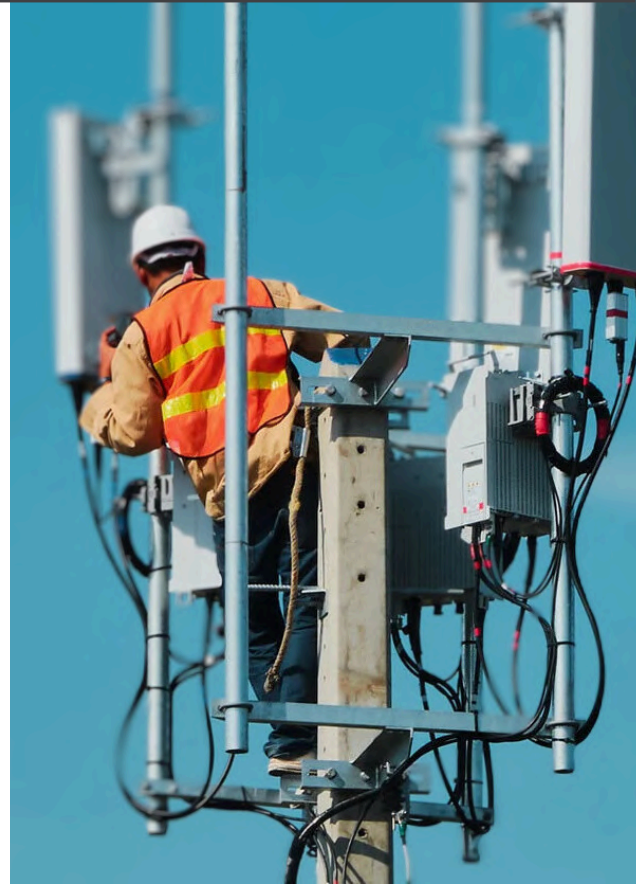
- ✓ **Antenna Height & Azimuth Optimization** – Adjusting antenna height and direction to enhance coverage and reduce overlapping signals.
- ✓ **Antenna Tilt Adjustments** – Fine-tuning electrical and mechanical tilts for optimal coverage and interference reduction.
- ✓ **Tower Load Balancing** – Ensuring equal distribution of network traffic across multiple towers to prevent congestion.

Logical Optimization

- ✓ **Parameter Tuning & Handover Optimization Analysis** – Analysing handover failure issues and reporting back to customer for adjusting handover thresholds and neighbor lists to ensure seamless call transitions between sites.
- ✓ **Carrier Aggregation & Frequency Optimization** – Monitoring and reporting LTE carrier aggregation parameters and frequency bands to maximize data throughput.
- ✓ **Interference Analysis & Reduction** – Identifying and mitigating co-channel and adjacent-channel interference to enhance network quality.
- ✓ **Quality of Service (QoS) Improvements** – Optimizing latency, packet loss, and throughput for better voice and data performance.
- ✓ **Actionable Insights & Recommendations** – Providing detailed recommendations for network improvements based on real-time field data.

Report Preparation & Data Analysis

- ✓ **Post-Processing of Drive Test Logs** – Analyzing log files collected from RF tools to extract key performance indicators (KPIs).
- ✓ **Coverage & Signal Quality Analysis** – Analyzing RSRP, SINR, RSRQ, throughput and other KPI's. Identifying areas with high call/PS/handover drops & failures, proposing optimization solutions.
- ✓ **Graphical Reports & Heatmaps** – Generating detailed reports on those KPI's across different test routes both before and after optimization as per customer's approved template, Visualizing network performance using Google Earth & MapInfo highlighting weak coverage areas.
- ✓ **Benchmarking Reports** – Preparing comparative reports against competitor networks, showcasing signal strength, latency, and data speed comparisons.



Optical Fiber services

we provide comprehensive optical fiber network solutions, including fiber laying, maintenance, end-user support, and customer connections. Our expertise ensures seamless high-speed connectivity for telecom operators, ISPs, and enterprises.



Our Services Include:

1) 🛠️ Optical Fiber Laying & Maintenance

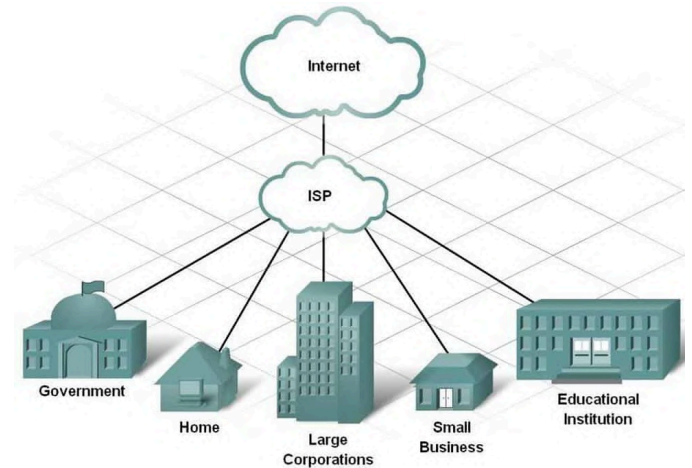
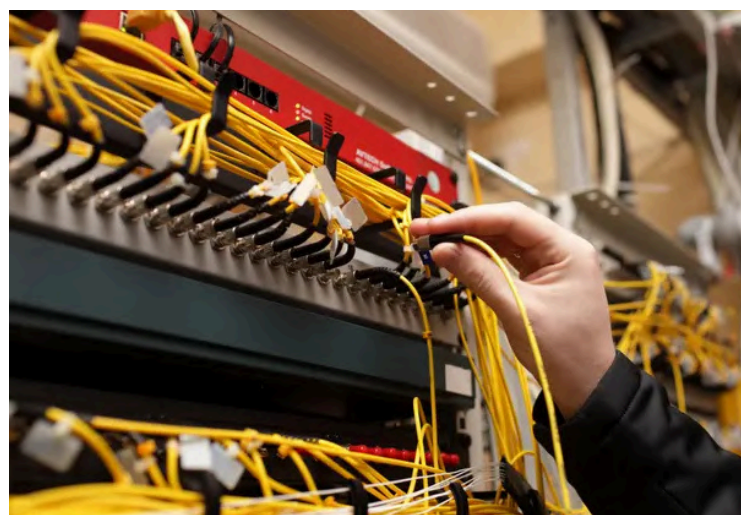
- ✓ **Underground & Aerial Fiber Laying** – Deploying fiber optic cables using both trenching, micro-trenching, and pole-mounted aerial methods.
- ✓ **Splicing & Termination** – Conducting high-precision fiber splicing, fusion splicing, and termination to ensure minimum signal loss.
- ✓ **OTDR Testing & Troubleshooting** – Using Optical Time-Domain Reflectometers (OTDR) and power meters to identify fiber cuts, attenuation issues, and connectivity problems.
- ✓ **Fiber Route Survey & Network Design** – Conducting site surveys, feasibility studies, and GIS mapping to design efficient fiber networks.
- ✓ **Preventive & Corrective Maintenance** – Regular inspections, cleaning, and repairs to ensure optimal fiber network performance and minimize downtime.



2) 🏢 POP Site Installation & Maintenance

- ✓ **POP Site Deployment** – Setting up regional distribution hubs for high-speed fiber connectivity.
- ✓ **Rack & Equipment Installation** – Installing switches, routers, OLTs (Optical Line Terminals), and power backup systems at POP sites.
- ✓ **Fiber Backhaul Management** – Connecting POP sites with the core network using fiber-optic backhaul solutions.
- ✓ **POP Site Security & Monitoring** – Implementing power backups, cooling systems, and remote monitoring for uninterrupted service.
- ✓ **Upgrades & Capacity Expansion** – Expanding fiber network capacity to support growing user demands and bandwidth requirements.

3) 🧑🏠 End-User Support & Customer Connections



- ✓ **New Fiber Connection Setup** – Installing fiber connections for homes, businesses, and enterprises, ensuring stable high-speed internet.
- ✓ **Home & Office Fiber Installation** – Providing end-to-end fiber setup, including ONT (Optical Network Terminal) configuration, Wi-Fi router setup, and LAN cabling.
- ✓ **Customer Issue Resolution** – Handling fiber-related complaints, connectivity troubleshooting, and service upgrades for end-users.
- ✓ **Bandwidth Testing & Network Optimization** – Ensuring customers receive optimal speed and performance through real-time network assessments.

Solar services

We utilize the latest solar technologies and techniques to ensure maximum energy output and long-term reliability. Our services include site assessments, permits, installation, and ongoing maintenance to keep your solar system running at peak performance. Our experienced team can design, install, and maintain high-efficiency solar panel systems to harness the power of the sun and reduce reliance on traditional energy sources for both commercial and residential clients.

☀️ Our Solar Installation Services

- **Solar Power Solutions for Telecom Towers & POP Sites**
- **Tailored for Enterprise & Commercial Needs**
- **Enhancing End-User Connectivity**



- ✓ **Solar Hybrid Power Solutions** – Integrating solar panels with battery storage and diesel generators for an efficient hybrid power system.
- ✓ **Off-Grid Solar Systems** – Deploying solar-powered energy solutions for telecom towers, POP sites as well as end user in remote locations.
- ✓ **Energy Optimization & Load Management** – Designing systems that optimize power consumption and improve energy efficiency.
- ✓ **Battery Storage & Backup** – Installing high-capacity lithium-ion battery banks for 24/7 power availability.
- ✓ **Maintenance & Support** - Routine inspections, cleaning, and repairs to maximize system lifespan, conducting solar efficiency tests, battery health analysis, and inverter diagnostics.

IT services

At Pie Square Technologies Pvt. Ltd., we offer comprehensive IT solutions tailored to meet the needs of businesses, government offices, telecom operators, and enterprises. Our services range from IT consulting and infrastructure setup to software development, cybersecurity, and web solutions. We help businesses streamline operations, enhance security, and drive digital transformation through cutting-edge technologies.

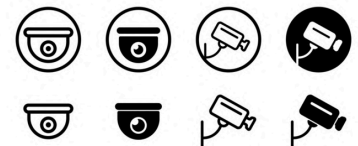
◆ We provide

IT Consulting & Infrastructure Solutions

We enhance your business's connectivity, scalability, and security through IT consulting and infrastructure solutions. Our expertise includes LAN, WAN, and hybrid cloud design and implementation. We provide 24/7 IT support and managed services for troubleshooting and maintenance.

We prioritize cybersecurity with firewalls, encryption, antivirus, and penetration testing to safeguard against threats. Our services create a resilient IT infrastructure for growth and operational needs.

CCTV Installation & Security Solutions



We provide a wide array of CCTV installation and security solutions, utilizing advanced technologies like IP and AI-enhanced cameras for superior image quality and analytics. Our systems enable remote access, allowing you to monitor your property from anywhere, ensuring your peace of mind. In addition, we offer specialized access control solutions to manage entry and enhance security for sensitive areas. To maintain optimal performance, our team delivers comprehensive maintenance services, ensuring your security systems remain effective. Whether upgrading or installing new systems, we tailor our solutions to meet your specific security needs.

Software, Web Development & Digital Solutions



We offer custom software and web development solutions tailored to business needs. Our services include enterprise software, ERP, CRM, and automation tools, as well as mobile app development for Android and iOS. We specialize in responsive website design, e-commerce platforms, and SEO-friendly web solutions, ensuring seamless user experiences. Additionally, we provide API development, system integration, cloud hosting, and cybersecurity measures to enhance performance, security, and scalability for businesses in various industries.

Project Expertise

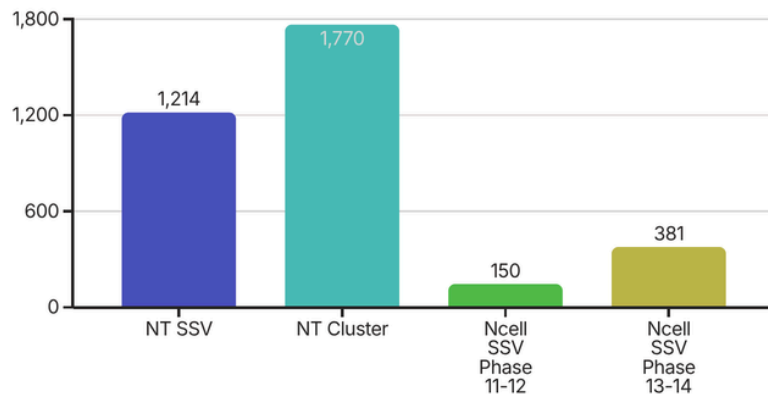
RF Drive Testing: 3500+ Sites

Extensive Coverage

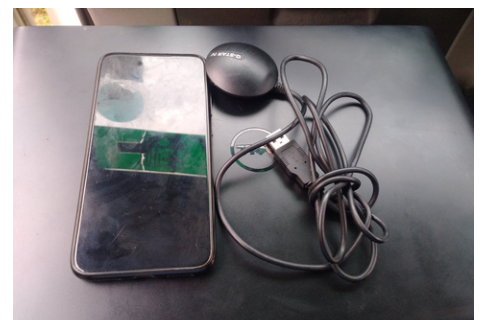
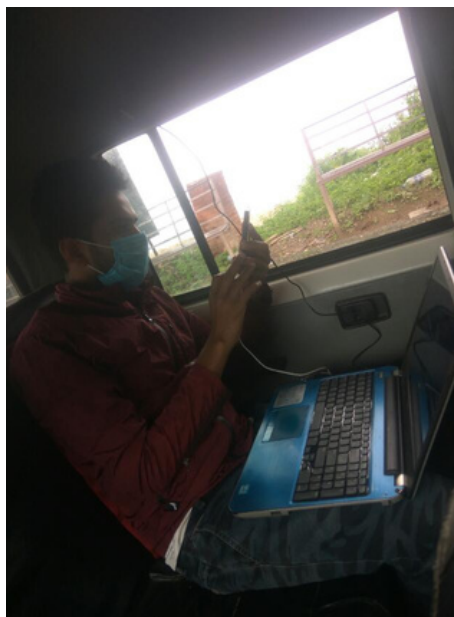
Proudly conducted RF drive tests for over 3500 sites of top telecom giants NTC and Ncell combined, covering different geographical areas across Nepal.

Optimization

We optimized network coverage and capacity. This enhanced user experience.



- Successfully executed Single Site Verification (SSV) drive tests for more than 1,214 sites as part of the NT-4G LTE project for Nepal Telecom.
- Conducted 66 cluster drives that included 1,770 sites under the NT-4G LTE project for Nepal Telecom, addressing more than 30 customer complaints.
- Completed SSV drive testing for over 150 sites within Phases 11 and 12 for Ncell.
- Completed SSV drive testing of 381 sites for Ncell that included 98 FTK SSV sites, 14 L900 sites, 8 L2100 sites, 243 4T4R sites, 15 multisector sites, 3 POC sites.



Project Expertise

Telecom Equipment Installation: 115 Sites

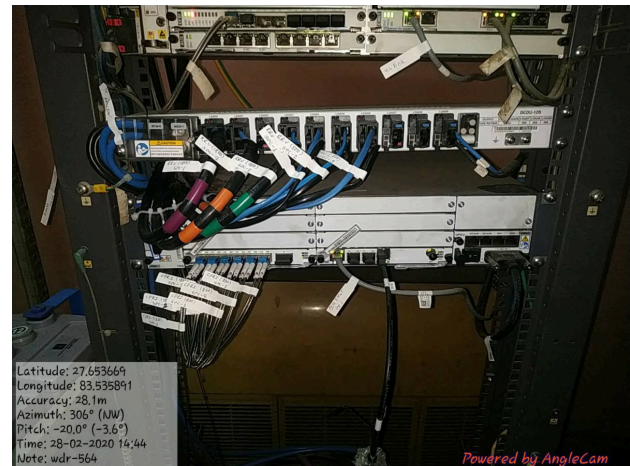
1 Expert Installation

Installed telecom equipment at 115 sites. Our team ensured seamless installation and commissioning. We followed industry's best practice.

2 Equipment Types

We handled a variety of equipment types, including base stations and transmission systems. We are proficient in the latest technologies.

- Successfully installed 57 new TI sites for Nepal Telecom as part of the NT-4G LTE project across the WDR region of Nepal, spanning from the Terai plains to the high hills.
- Completed the installation of 58 sites for Ncell in the KTM, WDR, and FWDR regions.
- Upgraded power from single-phase to three-phase at 152 sites for the NT-4G LTE project.



Project Expertise

Telecom Civil Works: 4 Sites

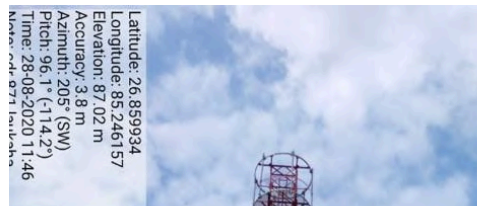
Infrastructure Development

Executed civil works for 4 telecom sites. This included site foundation construction, erection of tower, complete fencing and power connection.

Quality Assurance

We adhered to stringent quality standards. Our civil works projects ensured durability and reliability. We provide long-term value.

- Completed comprehensive telecom tower construction for 4 new sites as part of the NT-4G LTE project for Nepal Telecom. This included foundation construction, tower erection, Power connection, grounding, and fencing.



Project Expertise

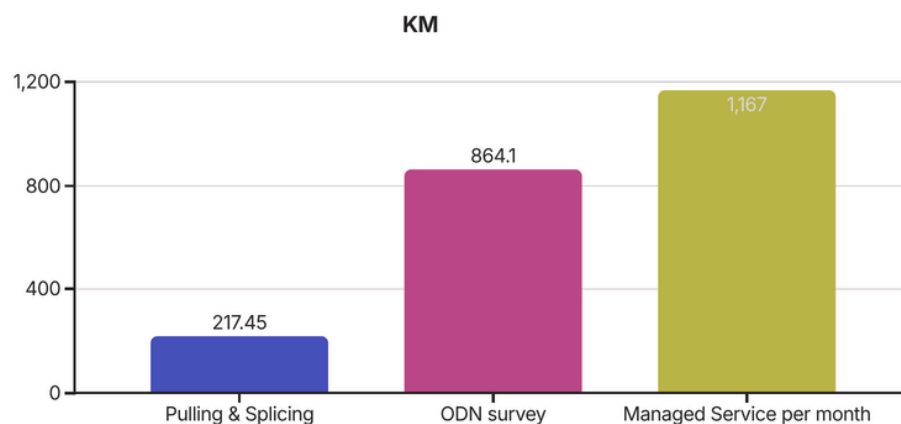
Optical Fiber: 2240+ km Survey, Installation & Maintenance



Surveyed, Installed and maintained over 2240 km of optical fiber. Our team provided customer connection to 9000+ users. Our services covered various locations throughout Nepal.



We ensured minimal downtime. Our maintenance services guarantee network availability. We respond promptly to issues.



- FTTH Pulling, Splicing and Installation of 217,453 meters for CG-net at locations including Ranibari, Pepsicola, Budhanilkanta, and Birgunj.
- Conducted a survey of 864,100 meters for the ODN ring for CG-net at Palpa, Kawasoti, Pragatinagar, Barghat, Manigram, Bharatpur, Nwalpur, Hetauda, and Ratnanagar.
- Surveyed 9 POP sites for CG-net located in Bharatpur, Nwalpur, Hetauda, and Ratnanagar.
- Established 1,351 customer connections in Kathmandu.
- Achieved 7650 customer connections outside the valley and ongoing.
- Provided managed service of an 1167 km network and 16 POP site support of CG-net, with ongoing maintenance till date.
- Installation, testing, and handover of the FTTH secondary network for 9 FDCs of Nepal Telecom.
- OFC cable pulling works in Rupandehi district for Nepal Telecom.

Project Expertise

Solar Installation & Maintenance



400 KW Solar O&M Project

We are providing comprehensive Annual Operation and Maintenance (O&M) services for a significant 400 KW ground-mount solar power plant, ensuring peak performance and reliable energy supply for our client.

Key Client Partnership

Proudly serving Surya Nepal Pvt. Ltd. in Simara, Bara, Nepal, demonstrating our expertise in large-scale solar projects.



As a subcontractor of Inventive Power and Consultancy Pvt. Ltd. in Satdobato, Lalitpur, Nepal, we delivered solar initiatives across multiple regions.

Solar Irrigation Systems

Implemented water pumping systems across **19 locations** in Madhesh and Lumbini Provinces (2081 B.S.). A key initiative under AEPC, enhancing agricultural productivity.

Municipal Solar Backup

Provided reliable solar backup systems for Mayadevi and Yasodhara Rural Municipalities in Kapilvastu (2080 B.S.), ensuring continuous operations for essential services.

Street Light Maintenance

Managed maintenance of solar street lights across Madhesh Province (2078 B.S.), in collaboration with the Ministry of Energy, ensuring public safety and energy efficiency.

Project Expertise

IT Consulting and Solutions

1

Provincial & Local Road Improvement Project (PPMU)

Delivered website page design, content addition, and updates, including DOIT template customization for PPMU, Janakpurdham Dhanusha, Nepal.

2

Purbadhar Bikash Nirdeshnalaya

Managed website content addition, updates on DOIT template, and provided staff training for Purbadhar Bikash Nirdeshnalaya, Janakpurdham Dhanusha, Nepal.

3

Nepal Health Research Council

Provided IT consultancy services, supporting crucial digital transformation initiatives to enhance research capabilities and data management.

4

National Reconstruction Authority

Implemented social media marketing, post boosting, page promotion, SEO, and developed media awareness content, alongside comprehensive social media account management and report generation.

Key Achievements

3500+

Telecom Sites Tested

Successfully completed RF drive tests.

115

Telecom Sites Installed

Telecom equipment installation milestones.

2240+

KM Fiber Installed

Extensive fiber optic network deployment.

4

Civil Sites

New site construction: foundations, tower erection, power, grounding, and fencing.

Solar Sites

400 KW AMC, Solar Irrigation & Backup, Streetlight Maintenance.

IT

Comprehensive IT consulting and digital transformation support.

These accomplishments reflect Pie Square Technologies' expertise and dedication to delivering high-quality solutions across telecommunications, IT, and renewable energy sectors, making a significant impact on Nepal's technological infrastructure. 🚀

Why Choose Us

Our Mission and Values

Our mission is delivering **high-quality, reliable, and innovative solutions** across **telecommunications, IT, fiber optics, renewable energy, and security infrastructure**. Our **expertise, efficiency, and customer-centric approach** set us apart, making us a trusted partner for businesses, telecom operators, and enterprises.

We are committed to integrity, excellence, and client success. We strive to exceed expectations in every project.



Integrity

We uphold the highest ethical standards.



Excellence

We strive for exceptional quality.



Client Success

Your goals are our priority.

Competitive Pricing and Value Proposition

We offer competitive pricing without compromising quality. Our solutions deliver exceptional value and ROI. We maximize your investment.

1

Quality

2

Value

3

Savings

Experienced and Dedicated Team

Our team brings diverse skills and deep industry knowledge. We are passionate about delivering exceptional results. Trust our experts to drive your success.

Innovative and Cutting-Edge Solutions

We embrace the latest technologies and methodologies. This allows us to provide forward-thinking solutions. Stay ahead of the curve with our expertise.

1

Latest Tech. Implementation

Implementation of new technologies.

2

Custom Solutions

Tailored to your specific requirements.

3

Future-Proofing

Ensuring long-term relevance and value.

Proven Track Record of Success

We have a strong industry portfolio across diverse sectors. We deliver exceptional results, on time and within budget. Our success speaks for itself.

Strong Industry Portfolio

Diverse projects, proven success.

Reliable Support

Consistent, responsive assistance.

Timely Completion

Delivered on schedule, every time.

Commitment to Ongoing Support

We provide reliable support and maintenance post-project completion. Our team is always available to assist you. We are dedicated to your long-term success.

1

Training

Comprehensive training sessions to empower our team.

2

Maintenance

Ongoing maintenance to ensure smooth operation.

3

Support

Responsive support team ready to assist you.

Personalized Approach to Client Needs

We listen, understand, and tailor solutions to fit your unique needs. Our collaborative approach ensures client satisfaction. We are committed to your goals.

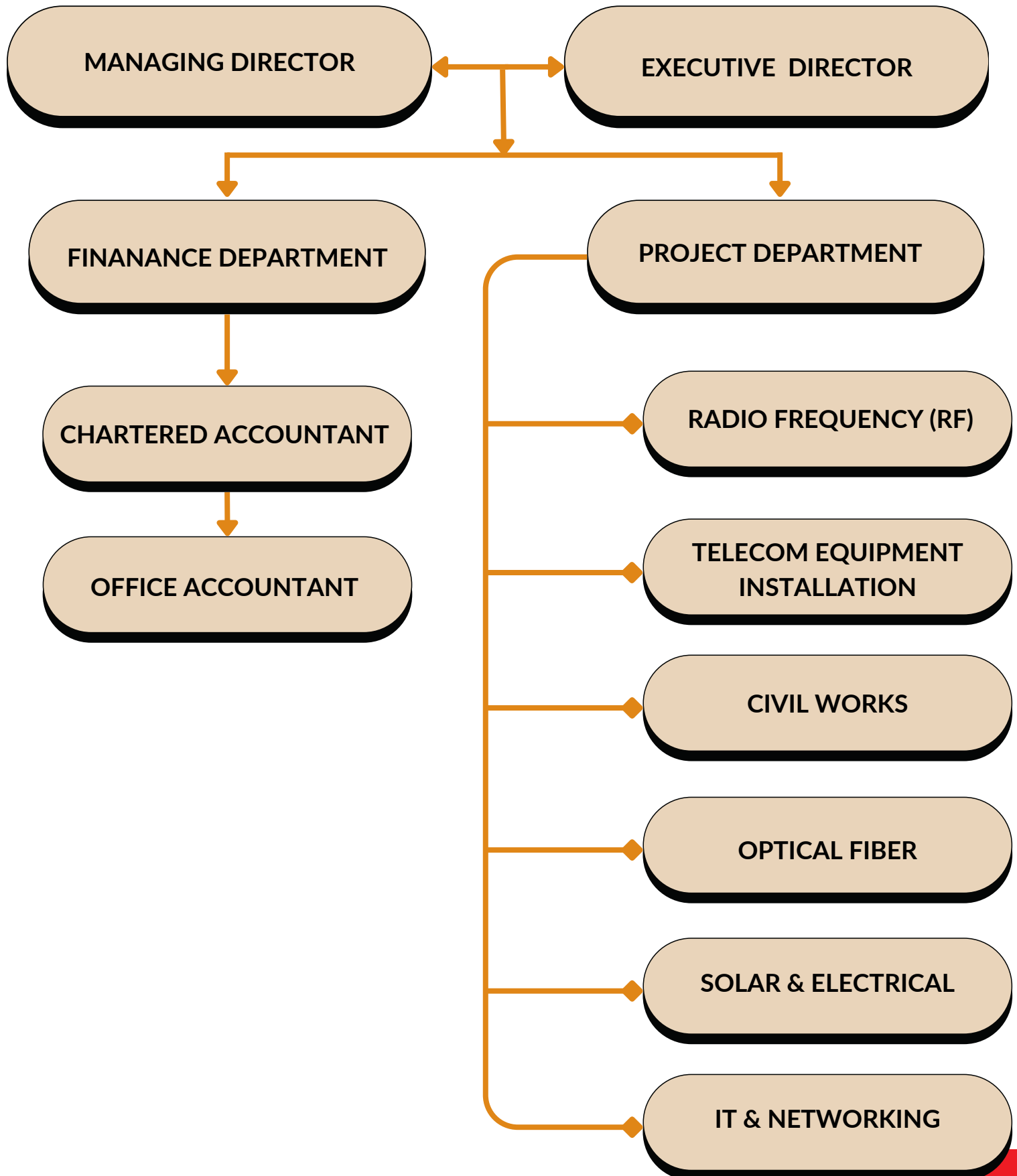
Dedicated Account Managers

Single point of contact for seamless communication.

Customized Project Plans

Tailored strategies to meet your objectives.

ORGANIZATION STRUCTURE OF THE COMPANY



Major Clients



中国通信服务
CHINA COMSERVICE



Nepal Telecom

CGNET

ZTE



SURYA NEPAL
PRIVATE LIMITED



PARAMOUNT
CONSTRUCTION
IMAGINE. PLAN. CONSTRUCT



Ncell

Contact us



pie square
TECHNOLOGIES
CONNECTING TODAY | BUILDING TOMORROW | POWERING FUTURE

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